Rethinking "Yellowstone in Yukon" and Baja British Columbia: Paleomagnetism of the Late Cretaceous Swede Dome stock, northern Canadian Cordillera

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[1] Paleomagnetic results obtained from the 69.8 Ma Swede Dome stock, emplaced into the pericratonic Yukon Tanana Terrane (YTT) west of Dawson, Yukon, indicate a minimal northward translation of 360 ± 575 km and a clockwise rotation of 20° ± 23° with respect to North America, in accord with the displacement of 425 km along the nearby Tintina Fault. Coeval Carmacks Group volcanics in contrast have previously indicated ~1900 km northward translation and minimal rotation and have been proposed to be a displaced manifestation of the fixed Yellowstone hot spot, thus providing an independent estimate of the Late Cretaceous paleoposition of the "Baja British Columbia" terranes of western North America. A compilation of recent mid and Late Cretaceous paleomagnetic results from the YTT and Intermontane Belt (IMB) terranes reveals that the Carmacks Group paleomagnetic result is anomalous. The YTT and IMB terranes, which form the bulk of Baja British Columbia, appear instead to have experienced <1000 km northward translation with respect to North America since 70 Ma, consistent with geological estimates of motion along plausible accommodating faults. In addition, past mobility of the Hawaii-Emperor hot spot implies that the Pacific hot spots may have experienced ~1200 km of southward motion from 81 to 47 Ma. If so, then the paleoposition of the Yellowstone hot spot if it existed at 70 Ma would likely have produced the Carmacks magmatism ~1000 km north of Yellowstone's current latitude, consistent with a "moderate" (<1000 km) post-70 Ma northward translation of the YTT and IMB terranes to their present position in North America.

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1. Introduction

[2] Paleomagnetic evidence for large-scale Mesozoic to Cenozoic terrane motions along the western margin of North America has fueled three decades of controversy. Suspect or accreted terranes comprising much of the Cordillera in British Columbia, Yukon and southern Alaska have, largely on the basis of paleomagnetic evidence, been proposed to have resided together as "Baja British Columbia" (Baja BC) some 2000–3000 km south of their present location in mid to Late Cretaceous time [*Irving*, 1985; *Irving and Wynne*, 1990]. Such mobility along the margin has been ascribed to the coupling of these terranes with the northward moving Kula oceanic plate between ~90 and 50 Ma [*Engebretson et al.*, 1985; *Irving et al.*, 1995; *Enkin et al.*, 2003], whereas geological evidence for such large terrane movements remains equivocal [*Gabrielse*, 1985; *Cowan et al.*, 1997; *Umhoefer*, 2000]. Much of the as-yet-unresolved debate concerns the timing, possible mechanisms and driving forces for such terrane movements, and also the validity of the paleomagnetic data and geological constraints [e.g., *Irving*, 1985; *Mahoney et al.*, 1999; *Housen and Beck*, 1999].

[3] A startling Late Cretaceous paleomagnetic result from the ~70 Ma Carmacks Group subaerial volcanics of the northern Cordillera found anomalously shallow paleomagnetic inclinations relative to those expected for North America, supporting an hypothesis for ~1900 km of poleward transport of the volcanics relative to ancestral North America

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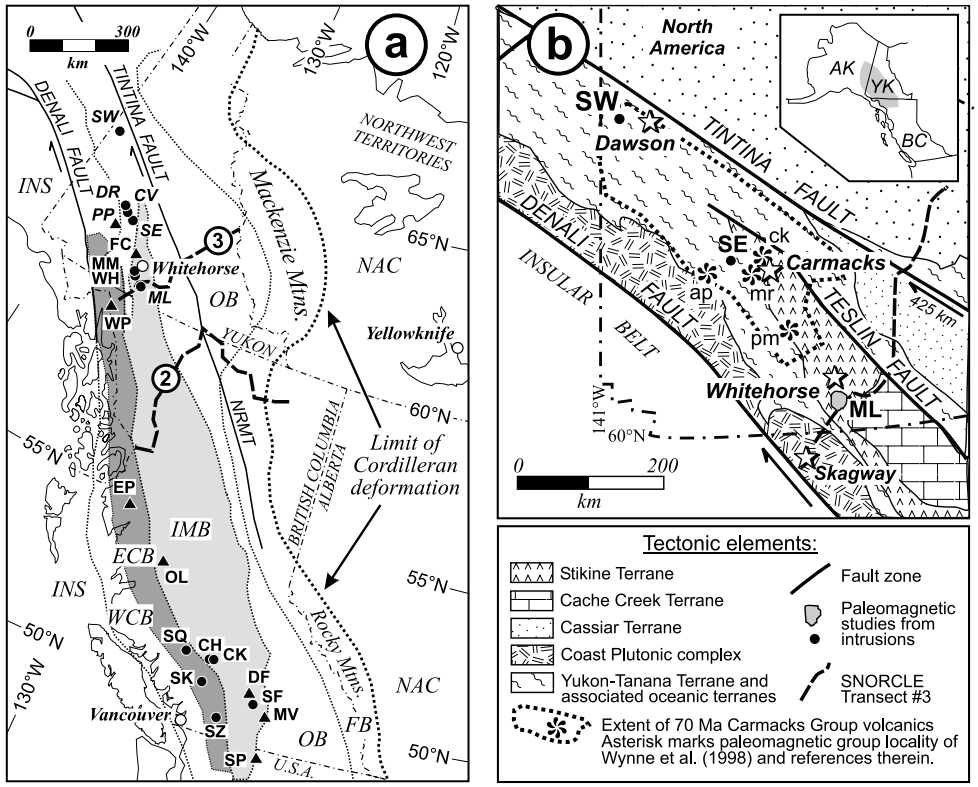


Figure 1. (a) Post-mid-Eocene morphogeological belts of the Canadian Cordillera [*Wheeler and McFeely*, 1991], abbreviated from east to west as FB, Foreland Belt; OB, Omineca Belt; IMB, Intermontane Belt; ECB, WCB, Eastern and Western Coast Belt; and INS, Insular Belt. The suspect or accreted terranes underlie much of the IMB, ECB, WCB, INS, and the Yukon Tanana Terrane (YTT) of south central Yukon. NRMT is the Northern Rocky Mountain Trench fault, an extension of the Tintina fault. LITHOPROBE-SNORCLE transects 2 and 3 are indicated by heavy dashed lines. Paleomagnetic study locations are plotted as mid to Late Cretaceous circles and Eocene triangles, with more detail given in Table 2. (b) Tectonic elements of the Northern Canadian Cordillera. The Stikine, Cache Creek terranes and portions of the Yukon Tanana Terrane underlie the Intermontane Belt. Inset abbreviations are AK, Alaska; BC, British Columbia; and YK, Yukon. Late Cretaceous paleomagnetic study abbreviations are ML, 75 Ma Mount Lorne pluton [*Harris et al.*, 1999]; SE, 69 Ma Seymour Creek stock [*McCausland et al.*, 2001]; SW, 70 Ma Swede Dome stock, this study. Paleomagnetic results from the Carmacks Group are derived from the locations marked with an asterisk: ap, Apex Mountain; ck, Carmacks region; mr, Millers Ridge; pm, Pilot Mountain [*Wynne et al.*, 1998].

since the Late Cretaceous (*Marquis and Globerman* [1988]; refined and expanded by *Johnston et al.* [1996] and *Wynne et al.* [1998]). The Carmacks Group was deposited as a widespread volcanic succession upon uplifted, eroded rocks of both the Stikine terrane of the Intermontane Belt (IMB) and on the surrounding Yukon Tanana terrane (YTT) of south central Yukon (Figures 1a and 1b). The region subsequently experienced post-Eocene uplift and erosion, leaving only remnant volcanic centers and exposed shallow level Eocene plutons [*Tempelman-Kluit*, 1974; *Hart*, 1995; *Johnston et al.*, 1996]. Recognition of the volcanics' shoshonitic, high MgO geochemical signature led to the elegant "Yellowstone in Yukon" proposition that the terranes had resided some 1900 km to the south in Late Cretaceous time atop the Yellowstone plume, thus also fixing their offshore longitudinal position with respect to North America [*Johnston et al.*, 1996].

[4] *Johnston* [2001] further used the 70 Ma paleoposition "fix" offered by the Yellowstone plume affinity and paleomagnetic data to define a paleogeography for the southernmost, IMB and YTT block of the hypothetical ribbon continent termed SAYBIA (Siberia-Alaska-Yukon-British Columbia), which he proposed to have assembled to form Alaska in the manner of a "terrane wreck" from 80 to 50 Ma. In his proposal, the ~2000 km northward translation of the IMB-YTT block would have been accommodated by the oroclinal folding of the more northerly blocks of SAYBIA as they became Alaska.

[5] The post-Cretaceous poleward transport of the YTT and IMB terranes and the overlying 70 Ma Carmacks Group volcanics is testable with paleomagnetic studies from other Upper Cretaceous units within the two terranes. Paleomagnetic work conducted under the aegis of the LITHOPROBE-SNORCLE (Slave Northern Cordillera Lithospheric Experiment) transect has produced results for the 75 Ma Mount Lorne stock of Stikine terrane in the northern IMB [*Harris et al.*, 1999] and the 69 Ma Seymour Creek stock of western YTT [*McCausland et al.*, 2001].

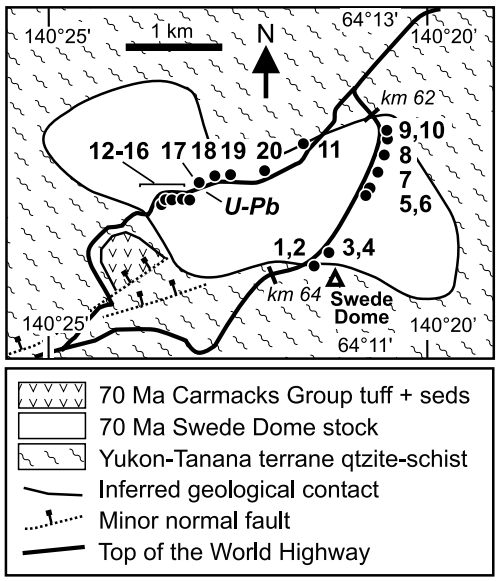


Figure 2. Geology and sampling sites for the Swede Dome stock, modified from *Mortensen* [1988] and *Gordey and Makepeace* [2000].

Both results imply lesser or no net transport of the terranes relative to North America since Late Cretaceous time. Here we present results from the 70 Ma Swede Dome stock in the northwestern YTT and reassess the "Yellowstone in Yukon" and SAYBIA hypotheses of Baja BC paleogeography in light of these new data.

2. Geological Setting

[6] The YTT is a poorly exposed assemblage of metamorphosed and deformed Paleozoic continental rocks which underlies a California-sized tract of south central Yukon to east central Alaska [*Mortensen*, 1992]. Its size, cratonic provenance, and "inboard" position with respect to the suspect terranes make the YTT a key tectonic element in the assembly of the northern Cordillera (Figure 1), but its past relations to other, apparently far traveled accreted terranes and to North America itself are poorly understood [*Mortensen*, 1992].

[7] By Late Cretaceous time, the pericratonic YTT is thought to have achieved its present configuration and adjacency to the volcanic arc/oceanic rocks of the Stikine, Cache Creek and Quesnel terranes to the south and east [*Tempelman-Kluit*, 1979; *Jackson et al.*, 1991; *Hart et al.*, 1995]. Extrusion of the 70 Ma Carmacks Group volcanics unconformably atop the inferred YTT/Stikine terrane boundary [*Tempelman-Kluit*, 1984; *Hart*, 1995] supports such a correlation. Post-Cretaceous dextral motion of the YTT and other accreted terranes with respect to North America along the Tintina Fault (Figure 1) has been estimated from offset of Cretaceous and older geological units at ~425 km [*Roddick*, 1967; *Tempelman-Kluit*, 1976; *Gabrielse*, 1985; *Colpron and Yukon Tanana Working Group*, 2001].

[8] West of Dawson, the YTT consists of Paleozoic greenschist to lower amphibolite facies schists and gneisses, with local occurrences of mostly thrust fault-associated greenstone, diabase and serpentinized harzburgite [*Mortensen*, 1988]. Late Cretaceous Carmacks Group andesitic flows and tuffs are essentially flat lying, undeformed and affected only by local small-scale normal faulting. The ovoid Swede Dome stock (Figure 2) outcrops over ~4 km^2 and has yielded a zircon and titanite U-Pb crystallization age of 69.8 ± 1.3 Ma [*Mortensen*, 1988], coeval with similar nearby intrusions and the widespread Carmacks-aged andesites, basalts and tuffs, dated at 70.2 ± 3.3 (simple average of five K-Ar and six Ar-Ar isochrons [*Smuk et al.*, 1997]). The stock was likely comagmatic with the Carmacks Group volcanic rocks, perhaps as part of a local volcanic centre for the tuffs and flows that are exposed 4 km to the southwest [*Mortensen*, 1988]. Regional aeromagnetic data compiled by *Gordey and Makepeace* [2000] show a local high that defines the extent of the stock.

[9] The Swede Dome stock (Figure 2) consists of a brown weathering, grey hornblende-biotite granodiorite to quartz monzonite and quartz syenite that typically has a 1–2 mm crystalline texture. In thin sections from 11 sites, samples have 35–50% plagioclase locally altered to clay minerals and carbonate, and 30–45% alkali feldspar, the latter occurring both in the matrix and as >2 mm megacrysts that exhibit a poikilitic texture, enveloping plagioclase. Up to 10% quartz is present and it is subhedral to weakly interstitial with plagioclase. Hornblende and biotite are each about 5%, forming the dominant mafic minerals. Hornblende is green to grey and locally altered to opaque Fe oxides. Biotite occurs as fresh, red-to-yellow pleochroic flakes. Accessory minerals are apatite, titanite, zircon and magnetite.

3. Paleomagnetism

[10] A total of 272 specimens were collected from 20 sites in the Swede Dome stock, along new and old sections of the Top-of-the-World Highway, 62 km west of Dawson (Figure 2). Each site is represented by at least six 2.54 cm diameter cores taken with a portable drill and oriented in situ by a magnetic and/or solar compass. Standard specimens of 2.20 cm length were sliced from each core and the bulk susceptibility of each was measured using a Sapphire Instruments SI-2B magnetic susceptibility meter. Specimens were then stored in a magnetically shielded room at the University of Windsor Paleomagnetic Laboratory, which has an ambient field of 0.2% of the Earth's geomagnetic field intensity, to allow some of their viscous remanent magnetization (VRM) to decay before measurement. All subsequent paleomagnetic and rock magnetic measurements were made in the shielded room.

[11] Specimens underwent stepwise demagnetization using thermal or alternating field methods to isolate their remanence characteristics. Measurements were performed using an automated Canadian Thin Films CTF DRM-420 cryogenic magnetometer, and demagnetization was done with a Sapphire Instruments SI-4 alternating field (AF) demagnetizer or a Magnetic Measurements MMTD-80 thermal demagnetizer. Four pilot specimens were demagnetized for each of the 20 sites, two specimens with 12 AF steps up to 130 mT and two specimens with 12–15 thermal steps up to 580°–645°C. Both AF and thermal demagnetization provided information that defined characteristic rem-

Table 1. Site-mean Bulk Magnetic and Remanence Data for the 70 Ma Swede Dome Stock, Yukon (64.2°N, 219.6°E)

Site	Susceptibility, $\times 10^3$SI	NRM, A/m	Site Type	ChRM Demagnetization AF, mT	ChRM Demagnetization TH, °C	n/N	R	In situ ChRM Direction D, deg	In situ ChRM Direction I, deg	α_{95}, deg	k
01	3.2	0.034	A	40–120	370–520	14/16		202.6	−75.0	3.0	173.0
02	4.1	0.042	A	40–120	450–560	14/14		194.3	−72.2	2.7	210.9
03	11.5	0.161	m	40–140	remag	14/14	7r	208.8	−76.7	2.4	277.3
04	6.9	0.053	m	40–150	500–570	8/12		141.8	−82.4	6.0	87.5
05	36.5	0.355	m	40–140	remag	9/13	8r	175.4	−83.8	3.8	185.4
06	44.7	0.381	B	40–120	remag	14/14	10r	143.8	−86.3	4.0	99.2
07[a]	44.1	2.010	B	60–130	450–600	11/13		247.1	−39.9	7.3	40.6
08	26.2	0.067	A	50–120	remag	10/14	6r	172.9	−81.4	2.7	315.5
09	24.5	0.094	m	30–120	450–580	12/13		179.9	−79.4	3.4	163.0
10[a]	31.9	0.336	B			0/17					
11	5.2	0.371	A	40–150	450–580	12/14		138.5	−77.9	2.4	332.1
12	36.2	0.420	B	50–150	450–560	9/11	7r	188.3	−81.4	4.4	138.2
13	37.6	0.164	B	remag	remag	6/11	6r	223.2	−82.6	5.7	138.3
14	36.8	0.151	B	30–120	remag	10/13	7r	102.8	−82.3	3.2	235.8
15	37.6	0.119	B	40–120	remag	8/11	5r	204.6	−79.4	4.9	131.6
16	36.6	0.110	B	40–130	450–560	12/13	7r	191.8	−88.6	2.9	230.0
17	24.5	0.124	m	30–150	450–520	11/13		255.4	−88.2	4.4	108.9
18	18.5	0.595	A	50–150	450–580	15/15		158.1	−82.4	2.8	185.4
19	12.5	0.573	A	40–150	450–600	15/15		147.3	−85.4	2.3	289.8
20[a]	16.0	1.520	A	50–150	450–600	16/16		294.5	−78.2	2.9	166.9
N = 17 sites								179.9	−82.7	2.8	164.8

[a]Sites 7, 10, and 20 are excluded from the collection mean calculation. Site 7 exhibits a non-Fisherian streak of specimen ChRM directions. Site 10 is dominated by a soft VRM such that its ChRM cannot be defined. Site 20 has a ChRM mean direction that is discrepant from the main collection mean, likely due to incomplete averaging of paleosecular variation. Susceptibility, site-mean bulk susceptibility; NRM, site-mean natural remanent magnetization intensity for specimens; ChRM demagnetization, typical range of demagnetization steps in alternating field (AF) and thermal (TH) over which the stable ChRM was isolated; n/N, number of specimens useable for site-mean ChRM calculation/total number of specimens in site; remag, number of stable (s) ChRM specimens that contribute to the site-mean plus number of specimens from which ChRM was determined using remagnetization (r) circles [*Halls*, 1976], typically defined from 15 to 120 mT and from 260°C to 500°C. Site-mean ChRM directions are given by D, declination; I, inclination; α_{95}, radius of 95% cone of confidence; and k, precision parameter estimate.

anent magnetization (ChRM) directions. Half of the remaining specimens were AF demagnetized in eight steps to 120 mT, and the rest of the specimens were thermally demagnetized in seven steps to 580°C. Many of the thermal demagnetizations were preceded by AF cleaning in three steps to 20 mT.

3.1. Rock Magnetic Properties

[12] Measurement of the Swede Dome specimens' natural remanent magnetization (NRM) showed a median NRM intensity of 16.7×10^{-2} A/m (first quartile, 7.5×10^{-2} A/m; third quartile, 52.0×10^{-2} A/m). Specimens from clinopyroxene-bearing sites 7 and 20 carried anomalously intense NRMs (median 169×10^{-2} A/m). The specimens had a median magnetic susceptibility of 28.1×10^{-3} SI volume units (first quartile, 15.0×10^{-3} SI; third quartile, 37.5×10^{-3} SI). Overall, both the NRM intensities and bulk susceptibility values found in the stock are typical for coarse grained felsic-to-intermediate intrusive rocks (Table 1).

[13] Thermal step demagnetization of two pilot specimens from each site provides information on the magnetic mineralogy and its domain state (Figure 3a). Neither goethite nor pyrrhotite appear to be significant remanence carriers, as there are no intensity drops or sharp directional changes associated with their diagnostic unblocking temperature ranges of 110°–150°C and 300°–350°C, respectively. Instead, specimens from the Swede Dome stock display sharp drops in intensity between 520°C and 560°C, diagnostic of magnetite as the remanence carrier.

[14] The 40 thermally demagnetized pilot specimens exhibit two distinct types of unblocking behavior. Group "A" specimens of Figure 3a show remanence intensities

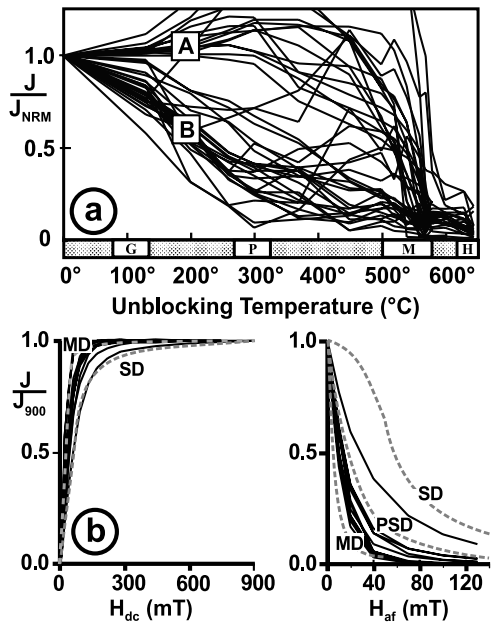


Figure 3. (a) Unblocking temperature spectra from thermal step demagnetization of 40 pilot specimens from the Swede Dome stock. Demagnetization behaviors "A" and "B" are discussed in text; G, P, M, and H refer to the diagnostic unblocking temperature ranges of goethite, pyrrhotite, magnetite and hematite, respectively. (b) Saturation isothermal remanent magnetization (SIRM) acquisition and decay curves for magnetite, indicating the presence of multidomain (MD) and pseudosingle-domain (PSD) magnetite remanence carriers in 14 representative specimens from the Swede Dome stock.

that unblock at high temperatures indicative of pseudosingle or single-domain magnetite, whereas group "B" specimens exhibit distributed unblocking temperature curves characteristic of a significant contribution from multidomain magnetite. Some specimens exhibit an apparent intensity growth during steps up to 450°C that is due to the progressive unveiling of an upward directed high-temperature component during the removal of the typically steep downward, lower-temperature component. Group "A" specimens are drawn from sites 1, 2, 8, 11 and 18–20, all from near the intrusion's contact with the host rock (Figure 2 and Table 1). Group "B" specimens represent sites 6, 7, 10, and 12–16, which are from the interior of the intrusion. Other sites have both types of unblocking behavior amongst their specimens. The geographic distribution of the A and B type of unblocking behaviors suggests that the cooling rate due to proximity of the intrusion's margin may have controlled the magnetite's effective grain size. This inference is supported for A-type sites 11, 18 and 19, which are finer grained in hand sample and thin section.

[15] Saturation isothermal remanent magnetization (SIRM) acquisition and decay curves were obtained for 14 specimens using a Sapphire Instruments SI-6 pulse magnetizer by magnetizing them in 14 direct field steps up to 900 mT and then demagnetizing them in seven AF steps up to 130 mT. The presence of pseudosingle to multidomain magnetite remanence carriers in specimens from the Swede Dome stock is confirmed by their SIRM behavior (Figure 3b).

[16] The presence of magnetite was also confirmed during a reconnaissance electron microprobe study of polished thin sections from four sites of the Swede Dome stock. Nonquantitative electron microprobe analysis of petrographic opaques identified magnetite and locally ilmenite. Magnetite occurs in association with hornblende as 5–50 μm grains, often in ~100 μm aggregate groups that are weakly subdivided by unknown silicate. The larger magnetite grains and aggregate groups are observed to enclose euhedral ~10 μm titanite laths. Small magnetite to titanomagnetite grains (<10 μm) occur within and immediately adjacent to hornblende, likely as a deuteric alteration. The large magnetite grains may host much of the multidomain remanence behavior, whereas the pseudosingle-domain remanence behavior may reside in the smaller magnetite grains within hornblende and perhaps in subdivisions of the larger magnetite grains.

3.2. Specimen Demagnetization

[17] In most specimens a significant down directed magnetization was removed by 450°C (Figure 4b) or 20 mT (Figure 4d), revealing an up-directed ChRM component. Pilot specimen demagnetizations showed this to be a mixture of a typically large, down-directed viscous remanent magnetization (VRM) that was removed by 10–15 mT or 260°C and another component that persisted to the higher steps. Only a small VRM was removed in specimens from the finer grained sites 11, 18, and 19 (Figures 4a and 4c). Together, these observations suggest that the magnetization removed by 20 mT or 450°C is a mixture of true VRM and a more persistent, normal polarity remanence carried by multidomain magnetite, which has been shown experimentally to be capable of carrying remanence acquired at ~200°C up to an unblocking temperature of 450°C [*Dunlop and Argyle*, 1991].

[18] ChRM remanence is recognized as an up-directed component resolved between 30 and 150 mT or 450° and 580°C. During AF demagnetization, the ChRM decayed toward the origin, removing most of the remaining remanence (Figure 4c). Thermal demagnetizations (Figures 4a and 4b) unblocked the ChRM with a sharp intensity drop from 520°C to 560°C. A ChRM vector was obtained over four or more consecutive AF or thermal steps between ~30 and 150 mT or 450°C and 580°C for each specimen by the least squares method of *Kirschvink* [1980], with a maximum uncertainty of <15° for the fitted line.

[19] In sites 3,5,6,8 and 12–16, the ChRM did not decay directly toward the origin on thermal demagnetization, but instead remained offset from the origin with a remanence that was only removed by the 645°C step, indicating hematite as the resilient remanence carrier. Since AF demagnetization of companion specimens shows no evidence for a hematite-borne remanence at high coercivities, the hematite must have formed and acquired its remanence in the oven. Close examination of thermal demagnetization plots at these sites shows that their ChRM vectors may be contaminated by an oven generated spurious remanence carried by hematite. Accordingly, their ChRM information was recovered using remagnetization circles from the 260°C to the 500°C steps [*Halls*, 1976; *Bailey and Halls*, 1984]. A few other thermal and AF specimens (Figure 4d) with unstable endpoints were also analyzed with remagnetization circles, typically defined from 260°C to 500°C and 15–120 mT. For specimens requiring the use of remagnetization circles, both the tendency for minor growth of hematite and the likelihood of VRM-dominated remanence directions appear to be related to their multidomain magnetite content, as indicated by SIRM testing and the larger mean specimen bulk susceptibilities at those sites.

3.3. Paleomagnetic Results

[20] Out of the 272 specimens in 20 sites that were demagnetized, 193 were deemed fit to provide ChRM information for 17 sites. Of these, 130 provided acceptable best fit lines for their ChRM decay paths, and remagnetization circles were used for 63 specimens. Site and collection mean ChRM directions (Table 1) were derived using *Fisher* [1953] statistics.

[21] The mean ChRM direction calculated for 17 sites has a declination of $D = 179.9°$ and inclination of $I = -82.7°$, with a radius for the cone of 95% confidence of $\alpha_{95} = 2.8°$ and a precision (directional dispersion) parameter estimate of $k = 164.8$ (Figure 5 and Table 1). Sites 7, 10 and 20 are excluded from the collection mean. Site 7 exhibits a non-Fisherian streak of specimen ChRM directions from southwest shallow-up to steep-up. Site 10 specimens are dominated by an intense soft VRM such that its mean ChRM cannot be defined. Site 20 specimens have large NRM intensities and demagnetization behaviors similar to those of site 7, with ChRM directions that are well clustered but discrepant from those of the 17 accepted sites (Table 1).

[22] The inclusion of remagnetization circle derived ChRM directions does not significantly bias the overall formation mean ChRM direction. The mean ChRM direction of mixed-method sites (mean $D = 186.2°$, $I = -83.7°$;

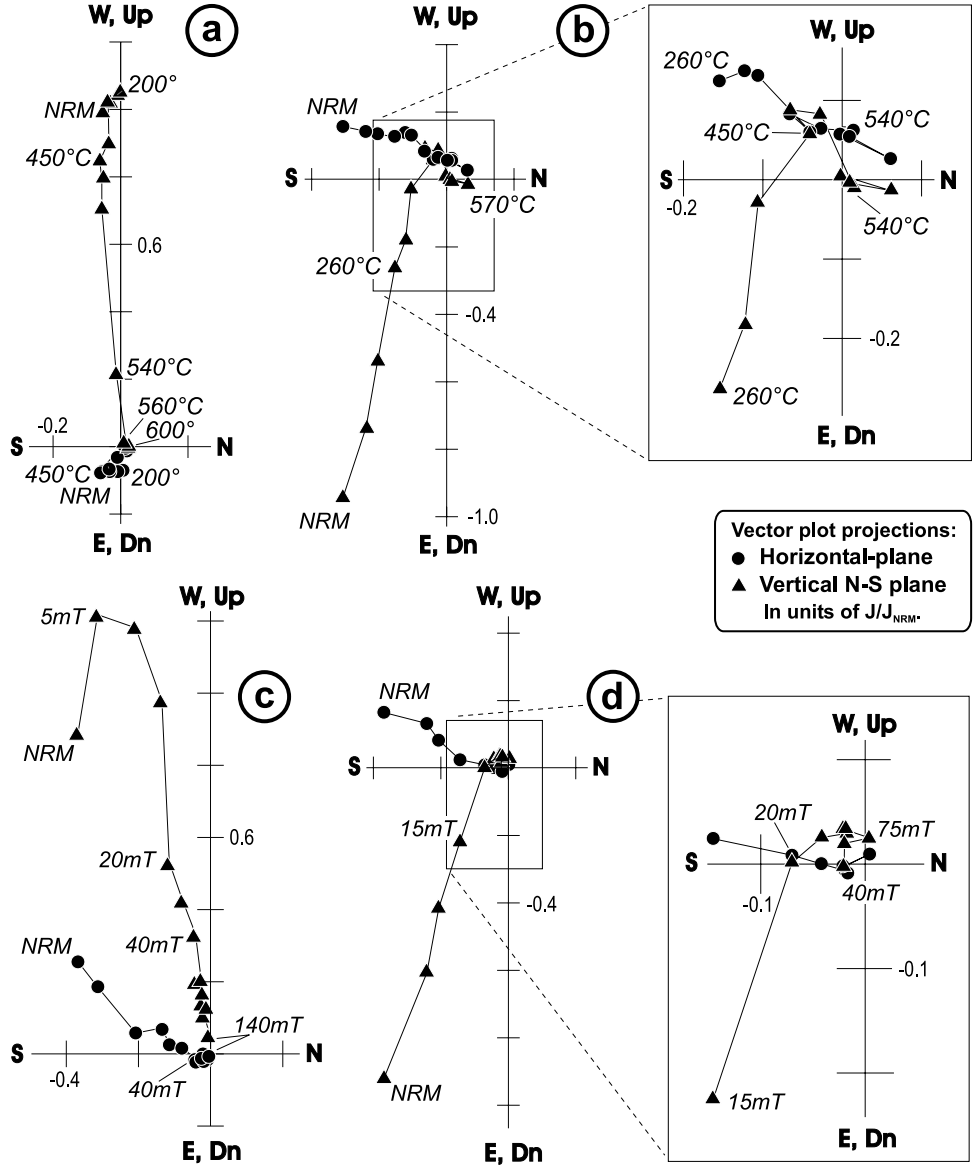


Figure 4. NRM intensity-normalized orthogonal demagnetization vector plots [*Zijderveld*, 1967] for four representative specimens of the Swede Dome stock. (a) Specimen 0601 from site 11 (NRM intensity is 2.6×10^{-1} A/m), showing clearly resolved thermally demagnetized ChRM directed steeply upward. (b) Thermal demagnetization of specimen 0401 from site 6 (NRM intensity is 3.0×10^{-1} A/m), in which VRM dominates with weak development of the reversed ChRM (inset) at high unblocking temperatures. (c) AF demagnetization of specimen 0601 from site 18 (NRM intensity is 4.2×10^{-1} A/m), with clearly resolved reversed ChRM. (d) AF demagnetization of specimen 0402 from site 6 (NRM intensity is 3.1×10^{-1} A/m), showing a VRM-dominated remanence similar to its companion specimen depicted in Figure 4b. See text for discussion. Circles (triangles) mark the stepwise demagnetization vectors projected into the horizontal W-E, N-S (vertical, N-S) plane.

α_{95} = 3.6° and k = 208, R = 8.96 for N = 9 sites) is indistinguishable (P = 0.05 [*McFadden and Lowes*, 1981]) from the ChRM mean of sites in which no remagnetization circles were used to obtain ChRM directions (mean D = 174.7°, I = −81.4°; α_{95} = 4.9° and k = 130, R = 7.95 for N = 8 sites).

3.4. Timing of Remanence Acquisition

[23] A primary thermoremanent origin for the Swede Dome stock's ChRM is likely because this is the youngest significant thermomagmatic event recognized in the area [*Mortensen*, 1988]. The 4 km^2 exposed size of stock and its epizonal subvolcanic emplacement imply that a primary remanence would have been acquired soon after its emplacement. The well resolved, steeply reversed ChRM directions were likely acquired within the 69.8 ± 1.3 Ma emplacement age of the stock, matching the chron 31R period of reversed Earth's magnetic field polarity [*Opdyke and Channell*, 1996]. A conductive cooling calculation [*Carslaw and Jaeger*, 1959], assuming an initial magma

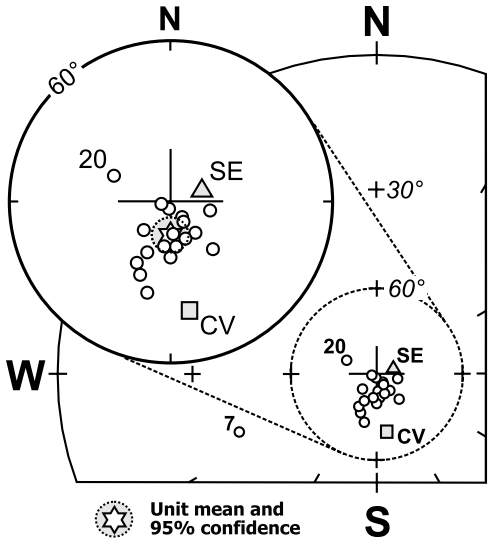


Figure 5. Equal-area stereographic projection upon the lower hemisphere of 19 reversed-polarity site-mean directions for the Swede Dome stock, along with reported reversed mean directions for the Seymour Creek stock (triangle, *McCausland et al.* [2001] and the Carmacks Group volcanics (square, *Wynne et al.* [1998]). Site 7 is excluded from the Swede Dome collection mean direction, as its specimen directions exhibit a non-Fisherian streak. For clarity, the inset circle shows the enlarged central portion (>60° inclination) of the projection, including the collection mean and 95% confidence ellipse for the 17 accepted Swede Dome sites.

temperature of 900°C and an ambient subvolcanic crustal temperature of 300°C, shows that the stock cooled through the characteristic magnetite blocking temperature range (~450°C) within 60 ka of the intrusive event, commensurate with the time required for paleosecular variation to have been averaged [*Merrill and McFadden*, 2003]. Sites near the margins of the stock will have cooled more quickly, and indeed the anomalous ChRM directions found in the fine grained marginal site 20 may represent a spot reading of the paleofield more akin to that of a volcanic flow.

4. Discussion

4.1. The 70 Ma Paleomagnetic Conundrum

[24] The mean ChRM direction of the Swede Dome stock defines a 70 Ma pole position at 78.6°N, 219.5°E (dp = 5.3°, dm = 5.5°, or A_{95} = 5.5°). When compared with the 70 Ma North American reference pole of *Besse and Courtillot* [2002] at 74.3°N, 205.2°E (A_{95} = 3.6°), the estimated net geotectonic motion of the Swede Dome stock with respect to North America is 3.2° ± 5.2° (360 ± 575 km) toward the reference pole, with a clockwise rotation of 20° ± 23° (Table 2 and Figure 6). This amount of northward motion is in accord with the estimated displacement of 425 km along the nearby Tintina Fault, which accommodated dextral motion between the YTT and North America from Late Cretaceous to Eocene time [*Murphy and Mortensen*, 2003].

[25] Paleomagnetic results from the coeval Seymour Creek stock [*McCausland et al.*, 2001] of the southern YTT also imply minimal net translation but give a large

Table 2. Mid-Cretaceous to Eocene Net Tectonic Motion Estimates From Units of the Intermontane Belt Terranes and the Yukon Tanana Terrane[a]

				Location		Paleopole			Reference Pole				
Unit	Name (References)[b]	Age, Ma	Sites	Lat, °N	Long, °E	Lat, °N	Long, °E	A_{95}, deg	Lat, °N,	Long, °E	A_{95}, deg	Net Translation Poleward,[c] deg	Rotation Clockwise, deg
WH	Whitehorse batholith (13)	112 ± 4	22	60.7	224.9	81.7	285.5	5.5	73.5	194.0	4.5	8.0 ± 5.7	41 ± 15
MM	Mount McIntyre pluton (12)	109 ± 1	11	60.7	224.9	70.5	299.6	5.8	75.5	194.4	5.8	11.9 ± 6.1	64 ± 16
SF	Spius Formation volcanics (9,10)	102 ± 2	32	50.1	239.9	63.7	312.4	6.6	76.3	195.3	6.0	8.0 ± 7.1	61 ± 12
CK	Churn Creek volcanics (10)	101 ± 4	49	51.4	237.6	57.6	299.2	5.7	76.3	195.4	6.2	5.3 ± 6.7	74 ± 13
SQ	Silverquick conglomerate (9,10)	97 ± 5	17	51.4	236.2	60.2	132.4	6.5	76.4	199.4	6.5	23.7 ± 7.0	−20 ± 12
DR[d,e]	Dawson Range batholith (11)	96 ± 4	13	62.6	221.6	68.7	208.0	10.9	76.4	198.2	6.4	−7.5 ± 10.1	17 ± xx
SZ	Spuzzum pluton (8)	96 ± 2	9	49.4	238.5	68.6	349.5	9.8	76.4	199.4	6.5	20.1 ± 9.4	43 ± 14
SK	Spetch Creek pluton (7)	94 ± 8	14	50.3	237.4	73.0	282.0	7.0	76.0	200.8	6.0	0.2 ± 7.4	42 ± 15
CH	Churn Creek conglomerate (5)	89 ± 6	16	51.3	237.3	73.2	039.4	4.5	75.4	207.4	5.2	28.0 ± 5.4	22 ± 10
ML	Mount Lorne pluton (6)	75 ± 3	18	60.5	225.3	78.3	290.9	4.3	74.6	206.7	5.2	11.1 ± 5.4	43 ± 18
CV	Carmacks Group volcanics (3, 4)	71 ± 2	26	62.1	222.8	78.4	088.6	7.8	74.4	205.6	4.0	23.1 ± 7.0	−6 ± 17
SW[d]	Swede Dome stock (this study)	70 ± 1	17	64.2	219.6	78.6	219.5	5.5	74.3	205.2	3.6	3.2 ± 5.2	20 ± 23
SE[d]	Seymour Creek stock (17)	69 ± 1	16	62.3	222.8	55.2	202.5	8.4	74.3	205.1	3.5	−0.9 ± 7.3	−94 ± 36
PP[d]	Pattison pluton (15)	56 ± 1	13	62.4	222.1	86.6	148.1	8.0	76.8	188.9	3.1	8.9 ± 6.9	17 ± 16
FC	Flat Creek pluton (19)	54 ± 1	21	60.9	224.5	82.2	171.4	3.9	77.5	182.6	3.3	3.8 ± 4.1	9 ± 10
MV	Marron volcanics (20)	52 ± 2	28	49.9	240.4	84.0	184.0	10.0	77.9	179.2	3.4	1.4 ± 8.4	10 ± 14
SP	Sanpoil volcanics (14)	52 ± 2	102	48.5	241.5	79.4	304.6	5.8	77.8	179.0	3.4	0.5 ± 5.4	34 ± 09
WP	White Pass dikes (1)	51 ± 1	24	59.4,	225.1	76.3	245.7	4.3	77.7	178.8	3.4	−5.3 ± 4.4	38 ± 13
OL	Ootsa Lake volcanics (2)	50 ± 2	13	53.7	233.2	88.0	354.6	11.5	77.5	178.2	3.4	6.7 ± 9.6	23 ± 16
DF	Dewdrop Flats volcanics (16)	49 ± 3	24	50.6	239.7	81.4	221.6	11.7	77.5	178.2	3.4	−3.6 ± 9.7	14 ± 19
EP	Eocene plutons (18)	46 ± 2	11	54.1	230.6	81.4	224.8	8.6	78.7	176.8	3.3	−3.1 ± 7.4	16 ± 16

[a]Calculated with respect to North America reference paleopoles from *Besse and Courtillot* [2002].

[b]Numbered references: 1, *Symons et al.* [2000]; 2, *Vandall and Palmer* [1990]; 3, *Johnston et al.* [1996]; 4, *Wynne et al.* [1998]; 5, *Enkin et al.* [2003]; 6, *Harris et al.* [1999]; 7, *Vandall* [1992]; 8, *Wynne et al.* [1995]; 9, *Wynne et al.* [1995]; 10, *Haskin et al.* [2003]; 11, *McCausland et al.* [2005]; 12, *Harris et al.* [1996]; 13. *Harris et al.* [1997]; 14, *Fox and Beck* [1985]; 15, *McCausland* [2002]; 16, *Symons and Wellings* [1989]; 17, *McCausland et al.* [2001]; 18, *Symons* [1977]; 19, *Symons et al.* [2004]; 20, *Bardoux and Irving* [1989].

[c]One degree of poleward transport is equal to ~111 km. For Cretaceous-Eocene time, translation toward the north pole (poleward transport) is equivalent to "northward transport" along the North American margin because the margin through that time was oriented along a north-south line.

[d]Units represent the YTT.

[e]Reported for the tilt-corrected result.

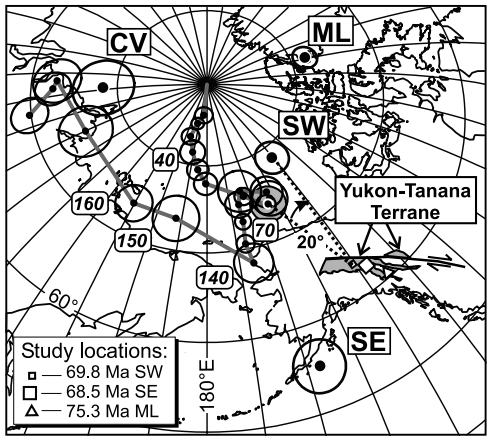


Figure 6. Late Cretaceous (75–69 Ma) YTT and Stikine Terrane paleopoles and their 1σ confidence ellipses plotted with reference to the synthetic North American apparent polar wander path of *Besse and Courtillot* [2002], with its 1σ bounds on mean poles at 10 Ma intervals. Paleopole abbreviations (see Figure 1b and Table 2) are CV, Carmacks Group; ML, Mount Lorne pluton; SE, Seymour Creek stock; SW, Swede Dome stock. See text for discussion.

counterclockwise (CCW) rotation estimate of 94° ± 36°. Results from the overlying but essentially coeval 70 Ma Carmacks Group volcanic rocks [*Wynne et al.*, 1998], however, imply a large poleward net translation for the YTT and Stikine terrane of 23.1° ± 7.0° (2550 ± 780 km) and minor counterclockwise rotation with reference to the same 70 Ma *Besse and Courtillot* [2002] North American pole (Table 2 and Figure 6).

[26] The translation estimate discrepancy between the Carmacks Group volcanics result and the intrusive results is perplexing, particularly because the two YTT intrusions are coeval and are likely to have been feeders for nearby Carmacks Group volcanic flows [*Mortensen*, 1988; *Johnston et al.*, 1996]! The discrepancies between all three coeval results cannot be due to terrane-scale geotectonic (paleolatitudinal or rotational) differences, but instead point to the presence of one or more erroneous assumptions in the analysis.

4.2. Possible Errors of Bias

[27] Within the Baja BC debate there has been considerable discussion of the veracity of paleomagnetic data from different rock types [e.g., *Irving et al.*, 1995; *Butler et al.*, 2001; *Symons et al.*, 2003], usually in regard to explaining the anomalously shallow Cretaceous paleoinclinations found in sedimentary rocks, plutons and volcanic rocks of the accreted Insular and IMB terranes. The geotectonic interpretation of anomalously shallow inclinations is that they represent the large northward translation of the host terranes relative to North America. Rotational discordance is usually observed to be clockwise [*Irving et al.*, 1995], and has been ascribed to terrane-scale rotation [e.g., *Harris et al.*, 1999], or to the corotation of smaller blocks [e.g., *Fox and Beck*, 1985]. *Butler et al.* [2001] noted that unrecognized tilting of intrusions and especially the effect of sediment compaction will tend to bias paleomagnetic directions to be shallower than expected. *Symons et al.* [2003] further discussed the possible causes of directional discordance in sedimentary, plutonic, and volcanic rocks. We will not reiterate those discussions in detail here, but call upon their salient points.

[28] As noted by many others [*Irving*, 1964; *Butler*, 1992; *Symons et al.*, 2003], the central issue in the tectonic interpretation of paleomagnetic data from any rock type is the exclusion of sources of directional bias, such as inclination shallowing due to postdepositional compaction of sedimentary rocks, unrecognized tilting of plutons, the insufficient averaging of paleosecular variation, and incomplete removal of a normal component (usually VRM) from reversed ChRM directions.

[29] For intrusions such as the 70 Ma Swede Dome and Seymour Creek stocks (Figure 1), it is important to refer the observed paleomagnetic directions to paleohorizontal. In both cases the nearby coeval Carmacks Group volcanic flows are generally flat lying, although few outcrops yield measurable surfaces and the shallow bedding orientations are typically difficult to ascertain [*Tempelman-Kluit*, 1974; *Mortensen*, 1988; *Marquis and Globerman*, 1988]. Exposure in both regions is poor, and minor syneruption or postemplacement tilting cannot be excluded [*McCausland*, 2002]. Most notably, the two coeval plutonic results are discrepant from one another in declination, which may either reflect relative tectonic rotation between the southern YTT and central YTT or an unrecognized tilting of one or both intrusions that is not reflected in the overlying volcanics.

[30] If the plutons' discordancies with respect to the 70 Ma North American reference direction were entirely due to postemplacement tilting, the tilt required would be 3° to 225° for Swede Dome, and 10° to 020° for Seymour Creek. Minor tilt is difficult to rule out, but more substantial tilt, such as would be required to match the shallower inclination Carmacks Group result is less likely. The Swede Dome stock has a better claim to paleohorizontal control in that the normal faults which cut the intrusion (Figure 2) hosted only minor vertical displacement [*Mortensen*, 1988]. The Seymour Creek stock is possibly cut by the Big Creek fault zone [*McCausland et al.*, 2001] which dextrally offset nearby 70 Ma Carmacks Group flows by 2 km [*Tempelman-Kluit*, 1974].

[31] For volcanic successions such as the 70 Ma Carmacks Group it is important to identify the number of independent cooling units amongst the sampled flows for successful averaging of secular variation (as debated by *Butler* [1990] and *Marquis et al.* [1990]). It is also important to determine if possible the contribution of paleoslope to the observed bedding orientations. The widespread Carmacks Group includes several thick volcanic units atop the YTT and Stikine terrane between Whitehorse and Dawson [*Tempelman-Kluit*, 1974] (Figure 1). The Group typically has a basal section of reversely magnetized, volcaniclastic andesite and andesitic flows with variable bedding orientations that were deposited disconformably on the preexisting uplifted basement topography. Up section, these fragmental andesitic units pass into generally horizontal basalt flows with a normal polarity magnetization at the stratigraphically highest levels [*Tempelman-Kluit*, 1974; *Marquis and Globerman*, 1988; *Johnston et al.*, 1996; *Wynne et al.*, 1998]. The paleopole reported for the 70 Ma Carmacks

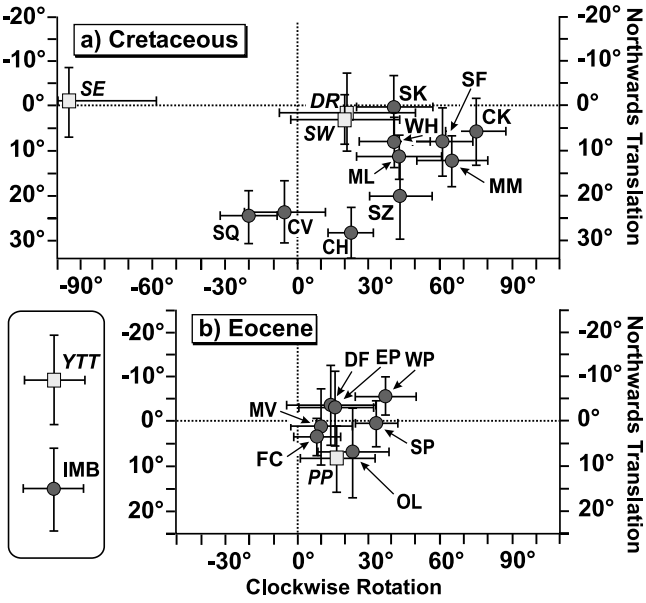


Figure 7. Paleomagnetic estimates of net northward translation relative to North America for units of the YTT and IMB terranes, Northern Cordillera, plotted against their net clockwise rotation for (a) mid to Late Cretaceous and (b) Paleocene–early Eocene time. Translation and rotation estimate uncertainties are derived [*Butler*, 1992] from the 95% confidence cones of the result paleopole and its coeval North American reference pole. Please refer to Table 2 for unit abbreviations, locations, and ages.

Group [*Johnston et al.*, 1996; *Wynne et al.*, 1998] is based on 26 tilt-corrected sites from three volcanic centers that are ~80 km apart: five sites from Pilot Mountain; 11 from the Carmacks region and Miller's Ridge; and 10 from Apex Mountain (Figure 1b).

[32] Comparison of site means between the upper, mostly horizontal and the lower, mostly andesitic flows reveals an inclination anomaly within the Carmacks result. The group consisting of dominantly flat-lying basalt gives ChRM directions whose mean is tightly clustered with an inclination that is 6.2° steeper than the mean direction of the more poorly clustered sites from the lower, variably oriented group. The upper basaltic group consists of five sites each from Pilot Mountain, Miller's Ridge, and Apex Mountain, comprising eight reversed and seven normal polarity sites [*Wynne et al.*, 1998; *Marquis and Globerman*, 1988]. These 15 sites provide a tilt corrected mean of $D = 159.4°$, $I = -71.5°$; $\alpha_{95} = 5.0°$ and $k = 58.5$. The remaining 11 sites drawn from the Carmacks region and the vicinity of Apex Mountain are all reversely magnetized and have eight separate bedding corrections [*Wynne et al.*, 1998; *Marquis and Globerman*, 1988] that give a tilt-corrected mean of $D = 173.9°$, $I = -65.6°$; $\alpha_{95} = 9.6°$ and $k = 23.6$. These variably oriented andesite and basalt sites contribute a shallowing bias to the overall Carmacks Group result and signal that there likely is a problem with their paleohorizontal control. We speculate that the bedding orientations of especially the lower, variably oriented sites contain an unrecognized paleoslope contribution that acts, even if randomly oriented, to bias steep directions to become shallower when falsely tilt corrected [cf. *Butler et al.*, 2001].

[33] In lieu of better paleohorizontal control for the Swede Dome and Seymour Creek stocks and a better estimate of the paleoslope contribution to coeval units of the Carmacks Group, the 70 Ma paleomagnetic conundrum appears to be irresolvable. Whatever the solution for these particular collections, the conundrum clearly illustrates a problem in relying on singular paleomagnetic results for the geotectonic analysis of terrane motions in the Cordillera: we cannot be absolutely sure that any particular paleomagnetic result is free of directional bias. It is therefore useful to obtain and compare similar aged Cordilleran paleomagnetic results from a variety of locations, rock types and structural settings to provide a collective estimate of terrane motions and a means of identifying anomalous results.

4.3. Toward a Collective Resolution

[34] Most relevant for comparison with the 70 Ma Carmacks Group volcanic and plutonic results discussed above is the paleomagnetism of the 75 Ma Mount Lorne stock [*Harris et al.*, 1999] of the IMB northern Stikine Terrane (Figure 1b), which gives an estimate of $11.1° \pm 5.4°$ (1230 ± 600 km) poleward translation and a clockwise rotation of $43° \pm 18°$ with respect to the 75 Ma reference pole for North America [*Besse and Courtillot*, 2002]. Paleohorizontal for the Mount Lorne stock is constrained with Al-in-hornblende geobarometry. The 75 Ma Mount Lorne result gives a poleward translation estimate that is interestingly intermediate between the >2000 km translation required by the Carmacks Group volcanic flows and the essentially minimal translation required by its two feeder stocks.

[35] A wider review of Cretaceous to Eocene paleomagnetic results from the YTT and IMB terranes, many of them obtained recently (Table 2), shows that the pattern of dominantly clockwise rotations noted by *Irving et al.* [1996] is still correct, but that the translation estimates exhibit a wide scatter from >2500 km northward motion to nil (Figures 7 and 8). Before considering the Cretaceous results and their implications for terrane paleogeography, it is useful to recognize that the group of Eocene net translation estimates (Figures 7b and 8) are centered on ~0° but scatter over ~13°, even though the accreted terranes are known to have been assembled near their present North American positions by mid-Eocene time [*Gabrielse*, 1985]. The eight Eocene paleomagnetic results, six of which include 0° translation in their 2σ error limits, come from a variety of rock types and tectonic settings including dikes, plutons, and volcanic flows (Table 2). Notice that no single, well resolved paleomagnetic result from the Eocene group provides a conclusive paleogeography, but collectively they do. The Eocene group also collectively shows that some clockwise rotation had yet to occur between 50 Ma and present, even though the terranes had arrived at 0° northward translation by 50 Ma [*Symons et al.*, 2003], thus implying that the suspect and accreted terranes have behaved as a coherent unit at least since Eocene time and likely beforehand.

[36] If the same consistency standard is applied to mid and Late Cretaceous results from the IMB and YTT terranes (Figures 7a and 8) it is apparent that there is a main group of eight clockwise rotated and moderately translated units, and

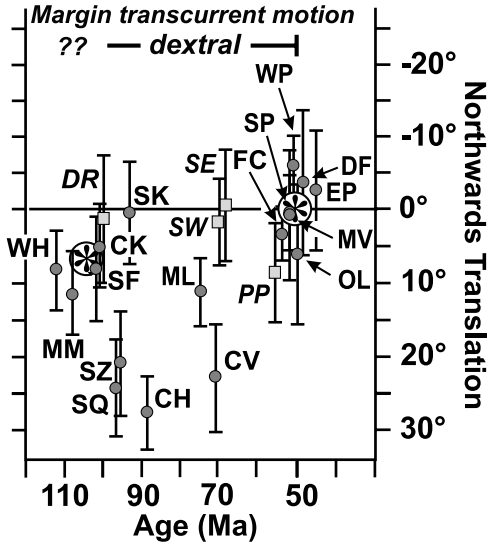


Figure 8. Cretaceous to Eocene paleomagnetic estimates of net northward translation for units of the YTT and IMB terranes relative to North America. Encircled asterisk symbols mark the geological estimate of northward translation for mid-Cretaceous and Eocene time [*Gabrielse*, 1985]. For clarity, uncertainties in age are not depicted but are given in Table 2. Uncertainty in northward translation is derived [*Butler*, 1992] from the 95% confidence cones of the result paleopole and its coeval North American reference pole. Please refer to Table 2 for unit abbreviations, locations, and ages.

a few anomalous results with more extreme inferred geotectonic motions. The outlier results are the Seymour Creek stock (with extreme counterclockwise rotation), and the Carmacks Group volcanics, the Silverquick and Churn Creek conglomerates (anomalously large northward translation and minimal rotation).

4.4. Assessment of Proposed Cretaceous to Eocene Paleogeographies

[37] Several versions of the far traveled Baja BC mid to Late Cretaceous paleogeography have been proposed on the basis of mostly paleomagnetic data from the Intermontane (IMB) and Insular Belt (INS) terranes [*Monger and Irving*, 1980; *Umhoefer et al.*, 1989; *Johnston*, 2001; *Enkin et al.*, 2003] and have been opposed on the basis of mostly geological data from these same terranes [*McClelland et al.*, 1992; *van der Heyden*, 1992; *Monger and Nockleberg*, 1996; *Mahoney et al.*, 1999; *Butler et al.*, 2001]. Here we will consider paleomagnetic results from the YTT and IMB terranes, which represent the bulk of the suspect and accreted terranes that comprise Baja BC, and from which many of the new paleomagnetic results are derived.

[38] The far southern location of the YTT and IMB terranes proposed in the SAYBIA paleogeographic model [*Johnston*, 2001] is based primarily on the paleomagnetic result from the Mount Tatlow Group [*Wynne et al.*, 1995] (now more specifically the Churn Creek conglomerate [*Enkin et al.*, 2003]) that indicates ~3000 km of northward motion relative to North America since mid-Cretaceous time (summarized by *Irving et al.* [1996] and *Enkin et al.* [2003]). The SAYBIA model incorporates the interpretation, based on geological and paleomagnetic arguments [*Monger and Price*, 1996], that the Insular and the IMB terranes were together by mid-Cretaceous time but disregards all mid and Late Cretaceous paleomagnetic results from the IMB terranes [e.g., *Irving et al.*, 1995; *Harris et al.*, 1996, 1997, 1999] other than that of the Carmacks Group [*Wynne et al.*, 1998].

[39] *Enkin et al.* [2003] further proposed that the combined Insular and IMB terranes may have undergone an extremely rapid southward translation of >20 cm/yr between ~102 Ma and 97 Ma, inferred from paleomagnetic results from the coeval 102 Ma Spius Formation and Churn Creek volcanics [*Irving et al.*, 1995; *Haskin et al.*, 2003] which provide a net northward translation estimate of ~1000 km, and a result from the stratigraphically overlying Churn Creek conglomerate (Table 2), correlated across the Insular-Intermontane boundary with the ~97 Ma Silverquick conglomerate portion of the Mount Tatlow Group, both providing a net northward translation estimate of ~3000 km.

[40] This proposed far southerly position for the Baja BC terranes finds support in the similarly large translation estimate from the nearby 96 Ma Spuzzum pluton [*Irving et al.*, 1985], but is at odds with the minimal translation estimate arising from the nearby, tilt corrected 94 Ma Spetch Creek pluton [*Vandall*, 1992]. *Brown and Burmester* [1991] performed geobarometry on the metamorphic rocks surrounding the Spuzzum pluton, and argued for a complex burial and uplift history that implies that the large translation estimate can be entirely attributed to subsequent southwest down 33° tilting of the pluton.

[41] The crux of the far traveled Baja BC interpretation remains the anomalous paleomagnetic results from the Mount Tatlow Group (Silverquick-Churn Creek conglomerate) and the Carmacks Group volcanics. These results imply large northward transport, but also display rotation that is distinctly anomalous from the significant clockwise rotations found amongst other IMB and YTT Cretaceous results (Figure 7a and Table 2). The Churn Creek and correlative Silverquick conglomerates may have suffered from inclination shallowing due to sedimentary compaction or an unrecognized contribution of a shallower remagnetization carried by authigenic magnetite [*Butler et al.*, 2001] (but see *Enkin et al.* [2003]). The rotational discordance for each of these results likely reflects local variation in declination between sampling locations that results from structural complexity in the sampled region (as recognized by *Enkin et al.* [2003] and *Haskin et al.* [2003]), although this possibility has not been explored for the Carmacks Group collection [*Wynne et al.*, 1998].

[42] We emphasize that collectively, mid-Cretaceous to Eocene paleomagnetic results from the YTT and IMB terranes are most consistent with moderate (~1000 km) northward translation of the terranes from ~105 Ma until 50 Ma (Figures 7 and 8), and do not support the proposed paleogeography and rapid geotectonic motions which place the YTT and IMB terranes 2000–3000 km to the south of their present position in mid-Late Cretaceous time [*Johnston*, 2001; *Enkin et al.*, 2003]. The inferred minimal mid-Cretaceous to Eocene geotectonic motions for the YTT and IMB terranes are in agreement with the suggested

paleogeography of these terranes in the "Andean arc" proposal of *Butler et al.* [2001]. Paleomagnetic results from the Insular Belt terranes, on the other hand, provide evidence for moderate to large mid and Late Cretaceous northward transport estimates [*Ague and Brandon*, 1996; *Irving et al.*, 1996; *Housen and Beck*, 1999], although a variety of alternative explanations have been proffered to explain their anomalously shallow inclinations [*Butler et al.*, 1989, 2001; *Kodama and Ward*, 2001].

5. Rethinking "Yellowstone in Yukon"

[43] A paleogeographic cornerstone of the "Yellowstone in Yukon" hypothesis is provided by attributing the interpreted geochemical plume affinity and uplifted eruptive setting of the 70 Ma Carmacks Group volcanics to their emplacement atop the inferred Late Cretaceous paleoposition for the Yellowstone hot spot [*Johnston et al.*, 1996]. Assuming hot spot fixity and the Late Cretaceous existence of the Yellowstone plume, the Yellowstone sourced Carmacks magmatism took place at 48°N, slightly offshore of the western margin of North America at 70 Ma, providing a Late Cretaceous paleolocation in latitude and longitude for the mobile Baja BC terranes.

[44] Recent paleomagnetic results from several of the Emperor seamounts reveal a striking mobility of the Hawaii hot spot prior to 47 Ma, and suggest that the Pacific hot spot group may have undergone ~1200 km of southward motion from 81 to 47 Ma [*Tarduno et al.*, 2003]. This is relevant to plate motion models [*Engebretson et al.*, 1985; *Kelley*, 1993] that use a fixed hot spot reference frame to estimate Kula-North American relative plate motions from the mid-Cretaceous to Eocene time. Much of the Kula plate's inferred rapid northward motion during this period may be accounted for by southward motion of the Pacific hot spots, thus eliminating the only plausible driving force for large-scale northward transport of the Baja BC terranes.

[45] A more particular implication of Pacific hot spot mobility prior to 47 Ma is that the 70 Ma paleoposition of the Yellowstone hot spot if it existed was likely ~1000 km north of its current latitude, leading to a critical modification of the "Yellowstone in Yukon" hypothesis. *Johnston et al.* [1996] suggested that the geochemistry and uplifted eruptive setting of the high-MgO Carmacks Group flows are consistent with the passage of a plume under the YTT and Stikine Terrane in Late Cretaceous time. If true, then a more northerly paleolocation of the Yellowstone hot spot in Late Cretaceous time is in better agreement with paleomagnetic and geological evidence for moderate ~1000 km post-70 Ma northward translation of the Baja BC terranes along the North American margin.

6. Conclusions

[46] Paleomagnetic results obtained from the 69.8 Ma Swede Dome stock, a feeder intrusion for the Carmacks Group emplaced into the pericratonic Yukon Tanana terrane (YTT) west of Dawson, Yukon indicate minimal net geotectonic motion of the YTT, with a northward translation of 360 ± 575 km and a clockwise rotation of 20° ± 23° with respect to North America. This amount of northward motion is reconcilable with the displacement of 425 km along the nearby Tintina Fault [*Murphy and Mortensen*, 2003]. The coeval and probably comagmatic Carmacks Group volcanics in contrast have given a paleomagnetic estimate of ~1900 km northward translation and minimal rotation. A compilation of recent mid and Late Cretaceous paleomagnetic results from the YTT and Intermontane Belt (IMB) terranes reveals that the Carmacks Group paleomagnetic result is anomalous. The YTT and IMB terranes comprising Baja BC appear instead to have experienced 1000 km or less net northward translation with respect to North America since 70 Ma, consistent with geological estimates of post-Cretaceous dextral motion along major faults between the terranes and North America. Past mobility of the Hawaii-Emperor hot spot implies that the Pacific hot spots may have experienced ~1200 km of southward motion from 81 to 47 Ma. The paleoposition of the Yellowstone hot spot at 70 Ma was likely ~1000 km to the north of its current latitude, consistent with a "moderate" (~1000 km) net northward translation of Baja BC.

[47] The far traveled Baja BC is an extraordinary hypothesis, requiring extraordinary evidence. Recognition of the pattern of moderate translations for the IMB terranes and the YTT, which together comprise the bulk of the proposed Baja BC crustal block, negates the strongest, i.e., paleomagnetic evidence for Baja BC that comes from a lesser number of studies mostly sampled from the western margin of the amalgamated Baja BC terranes. This newfound agreement between the paleomagnetic and geological estimates of coherent geotectonic motion for the YTT and IMB terranes cannot be understated. Paleomagnetic and geological data from the YTT and IMB terranes together point to a new understanding of the motions involved in the assembly of the Canadian Cordillera.

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